
SIMPLE ALGORITHM OF SLIDING AVERAGES WEB APP

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ABSTRACT

I describe a new web-based application for a simple algorithm, originally published in 2007. New additional features are incorporated to enable a user to better perform experiments using the algorithm. Some additional insights and discussion on the design and implementation of the algorithm itself are provided here and in a separate manuscript. Results of using the application for different protein properties are also presented in separate manuscripts.

Keywords sequence analysis · bioinformatics

1 Introduction

A Simple ALgorithm of Sliding Averages (SALSA) was originally designed for the purpose of estimating the propensity for cross- β -sheets found in amyloid fibrils for specific natively unfolded proteins. It uses a sliding window computation in which the property of interest is calculated for each window as an average over that window. Although it is a very straightforward sampling algorithm, its method includes unique aspects which, by their combination, appear to contribute insights into otherwise established characteristics of a protein, using only its amino acid sequence. Although originally designed for protein sequences, the web application places no restrictions on the type of sequence or property inputs SALSA will take, biological or otherwise.

2 Implementation details

The implementation was originally written in Java by the second author (O.S. Makin) of the original publication Zibae et al. [2007]. I wrote a new implementation in Python (https://github.com/zibae-2021/SALSA_mprops). The web application presented here was built with an agentic coding system in Python, Javascript, HTML and CSS. (The application is currently hosted on the free tier of Render). The code is not being made publicly accessible but the the application itself is free to use for academia.

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2.1 Algorithm design

The following step-by-step description of SALSA's computation aligns with the step-by-step decisions made during its design and implementation:

*Use footnote for providing further information about author (webpage, alternative address)—*not* for acknowledging funding agencies.

1. A property of interest is calculated as a simple mean over a short contiguous stretch of the sequence.
2. This calculation is repeated on overlapping windows, sliding along the sequence one position at-a-time, akin to the long-established Kyte and Doolittle protein hydropathy algorithm Kyte and Doolittle [1982].
3. Unlike the Kyte and Doolittle method, the score of a window is then assigned to every position in the window, not just the central one.
4. The sliding window calculation is repeated from the start, using a different size of window. The typical use-case is to employ about 10 or more different window sizes.
5. The batch of scored windows represents the main output of SALSA. It was not displayed in the original publication, but is being made available in the new web application - displayed in 'Windows Plot'.
6. To aggregate the many scored windows into a more understandable and quantifiable summary format, the scores are summed as follows:
 - (a) Lower-scoring windows are filtered out according to a user-defined threshold.
 - (b) The sum of scores at each position is taken across every window covering that position. As such, for a sequence of length 100, the result is a list of 100 summed scores, each one corresponding to one specific position in the sequence.
 - (c) The numbers can be plot against the sequence - displayed in 'Profile Plot'. This is second output.
7. A final summary score in the form of the sum of the summed scores is the third output. As such, this is a single number, regardless of sequence size.

My initial intention was that the scored windows output (as in 'Windows Plot') should be summarised in a way that does not lose information. However, I did not solve this seemingly impossible challenge. In the meantime the second author (O.S. Makin), who was also responsible for writing the original Java-based implementation of SALSA, came up with the summary computation (number 6 in the list above), which is displayed in Profile Plot. This turned out to be a very useful summary output and was crucial for the applying the algorithm in related studies.

$$\xi_{ij}(t) = P(x_t = i, x_{t+1} = j | y, v, w; \theta) = \frac{\alpha_i(t) a_{ij}^{w_t} \beta_j(t+1) b_j^{v_{t+1}}(y_{t+1})}{\sum_{i=1}^N \sum_{j=1}^N \alpha_i(t) a_{ij}^{w_t} \beta_j(t+1) b_j^{v_{t+1}}(y_{t+1})} \quad (1)$$

2.1.1 Headings: third level

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3 Examples of citations, figures, tables, references

3.1 Citations

Citations use natbib. The documentation may be found at

<http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf>

Here is an example usage of the two main commands (citet and citep): Some people thought a thing [??] but other people thought something else [?]. Many people have speculated that if we knew exactly why ? thought this. . .

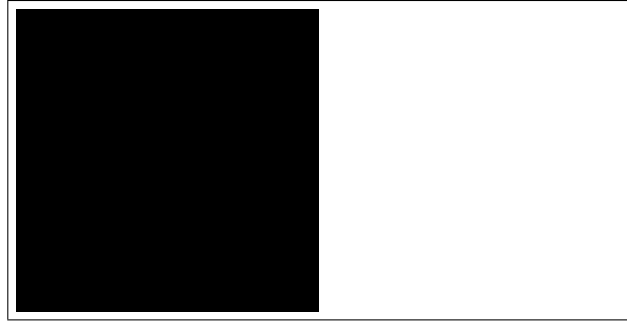


Figure 1: Sample figure caption.

Table 1: Sample table title

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	~ 100
Axon	Output terminal	~ 10
Soma	Cell body	up to 10^6

3.2 Figures

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3.3 Tables

See awesome Table 1.

The documentation for booktabs (‘Publication quality tables in LaTeX’) is available from:

<https://www.ctan.org/pkg/booktabs>

3.4 Lists

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References

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²Sample of the first footnote.